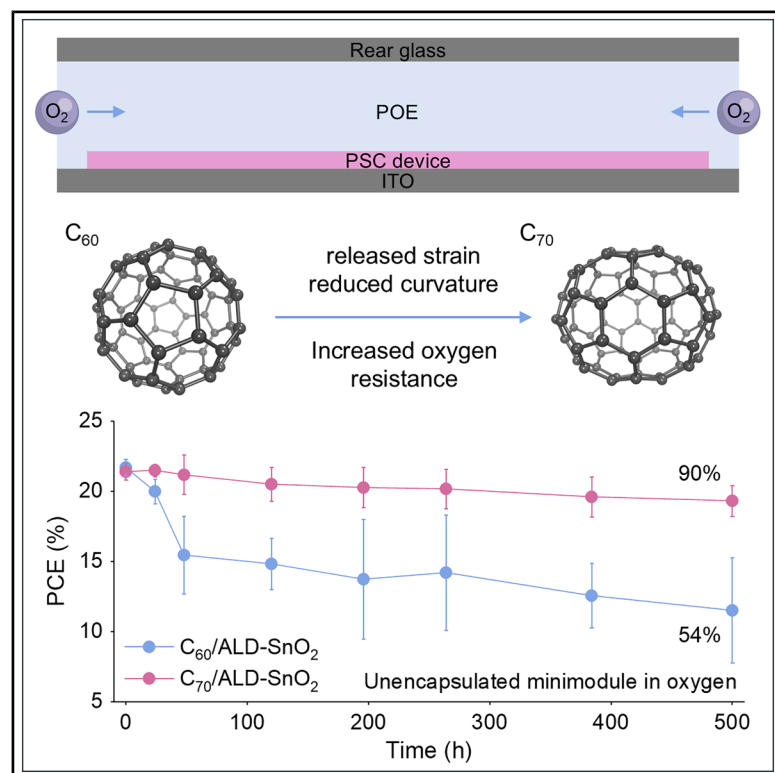


Elongated fullerene unlocks oxygen-resistant perovskite solar minimodules

Graphical abstract



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In brief

Existing encapsulation cannot prevent oxygen ingress in perovskite solar modules, leading to rapid degradation. We show that oxidation of C_{60} , rather than the perovskite, causes the failure of solar cells with C_{60} as the electron transport layer. Replacing C_{60} with strain-relieved C_{70} , together with a dense ALD- SnO_2 buffer, suppresses oxygen reactivity and diffusion, respectively. This strategy yields orders-of-magnitude stability gains and enables extrapolated module lifetimes to several decades, offering a practical pathway toward durable perovskite photovoltaics.

Highlights

- Oxygen ingress into encapsulated PSC modules in weeks or months
- Oxidation of C_{60} dominates the degradation of p-i-n structured PSCs
- Elongated fullerene C_{70} relieves structural strain and enhances oxygen resistance

Article

Elongated fullerene unlocks oxygen-resistant perovskite solar minimodules

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CONTEXT & SCALE We discovered that perovskite solar modules become saturated with oxygen within weeks to months in the field with state-of-the-art encapsulation, leading to device degradation. Instead of the perovskite absorber, we identified the C₆₀-based electron transport layer as the dominant failure bottleneck in the most common device stacking. Specifically, C₆₀ undergoes oxidation upon oxygen exposure, significantly impairing its electron transport properties. To address this limitation, we found that replacing C₆₀ with the elongated fullerene C₇₀, which has less structural strain, resulted in orders-of-magnitude device stability toward oxidation. In addition, a compact SnO₂ layer on C₆₀ formed by atomic layer deposition further suppresses oxygen diffusion. This combined strategy enables unencapsulated PSCs to retain 90% of their initial efficiency (T90) after 1,200 h of operation under 1-sun illumination in pure oxygen at 70°C—representing a stability improvement of several hundred times compared with devices based on C₆₀. This stability can translate to over 40 years of stability toward oxygen at 50°C.

SUMMARY

State-of-the-art encapsulation cannot prevent the permeation of oxygen into perovskite solar cells (PSCs). Here, we report the finding that p-i-n structured PSCs degrade quickly under oxygen exposure. Oxidation of C₆₀, rather than perovskites, dominates the degradation. Chemical absorption of oxygen by C₆₀ is identified, which impairs its electron transport property. We modify the electron transport layers to address the instability under oxygen exposure. The elongated fullerene C₇₀ is found to react with oxygen orders of magnitude slower than C₆₀. In combination with a compact SnO₂ buffer fabricated by atomic layer deposition, which can slow down the oxygen diffusion, the resulting unencapsulated PSCs with C₇₀ retained 90% of their initial efficiency after 1-sun illumination in pure oxygen for 1,200 h at 70°C, improving stability by hundreds of times. Testing of unencapsulated perovskite minimodules of different perovskite compositions with C₇₀ gives extrapolated lifetimes of 17–41 years at 50°C.

INTRODUCTION

P-i-n structured perovskite solar cells (PSCs) are currently dominating the commercialization of this technology due to their rapidly improving power conversion efficiency (PCE) and particularly their superior stability.^{1–3} The efficiency of small-area p-i-n PSCs has already exceeded 27%, and the perovskite minimodules with an area >100 cm² have exceeded 22%.⁴ T90 lifetime (the duration taken to reach 90% of the initial or peak efficiency) of over 1,000 h has been frequently re-

ported at 85°C and 1-sun illumination.^{5–7} However, translating accelerated indoor tests into reliable outdoor operational lifetimes is non-trivial, and such a relationship has not yet been established for PSCs. To enable perovskite-based technology to be commercially viable, outdoor operational stability needs to be demonstrated and has therefore become a focus for both academia and industry.⁸ To date, there have been no successful demonstrations of T90 lifetime over 2 years from public data for modules with an area larger than 50 cm² with over 15% PCE. This raises one important question: what causes the

difference between outdoor stability tests and indoor tests performed in an inert environment?

Multiple aspects are being studied by the community to understand the differences between indoor and outdoor stability tests, including ultraviolet (UV) light, light cycling, thermal cycling, shading, etc.^{9–11} UV light accelerates PSC degradation by damaging the interface of the hole transport layer and the perovskite.^{10,12} Light cycling may slow PSCs degradation compared with continuous illumination due to the nighttime self-recovery.¹³ Conversely, repeated stress cycling can activate fatigue that accelerates degradation.¹⁴ Many small-area cells or minimodules have been shown to pass the thermal cycling test, and thus, it might not be the main differentiator.¹⁵ Some early studies showed that a combination of oxygen and light could cause relatively fast degradation of perovskites.^{16,17} However, a common perception is that encapsulation should exclude oxygen outside perovskite modules. Recently, a study by Repins et al. challenged this perception.¹⁸ Their calculation based on the reported diffusivity of oxygen through common encapsulation materials, desiccated edge seal polyisobutylene (PIB), showed that they could only block oxygen for months or less, depending on the environmental temperature. To ensure that encapsulated perovskite modules can operate reliably outdoors for 25 years, it is crucial to address the concern of oxygen-induced instability.

In this work, we experimentally verified the rapid diffusion of oxygen through a desiccated PIB edge seal and a polyolefin elastomer (POE) based polymer encapsulant using an oxygen sensor embedded in mock perovskite minimodules. We found that unencapsulated PSCs with C_{60} as the electron transport layer (ETL) degraded to T90 within less than half an hour in pure oxygen and at 70°C. The degradation was primarily caused by the oxidation of C_{60} , rather than perovskites. The oxidation of C_{60} followed an Arrhenius relationship with a low activation energy of 0.37 eV. We found that elongated fullerene C_{70} could dramatically slow down oxidation by 116 times at 70°C with an activation energy of 0.61 eV. In addition, tin oxide (SnO_2) deposited by atomic layer deposition (ALD) was also found to effectively slow down oxygen penetration. Combining them increased the T90 lifetime of the unencapsulated PSC devices in pure oxygen to >1,200 h under 1-sun illumination with 4.5% UV light at 70°C. The T90 lifetimes ranging from 17 to 41 years at 50°C were extrapolated for the encapsulated perovskite minimodule based on the oxygen stability of the unencapsulated perovskite solar minimodule.

RESULTS

Unexpected degradation of perovskite minimodules in outdoor test

First, the PSC device with a configuration of tin-doped indium oxide (ITO)/EtCz3EPA/PTAA/FA_{0.9}CS_{0.1}PbI₃/C₆₀/BCP/Cu (Figure 1A) was fabricated, in which EtCz3EPA is [2-(9-ethyl-9H-carbazol-3-yl)ethyl]phosphonic acid, PTAA is poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine], FA is formamidinium, and BCP is bathocuproine. We previously showed that this device architecture can reach T90 lifetimes of >1,780 h under 1-sun illumination with 3.5% UV light at 60 ± 5°C in N₂.¹⁰ Thus, by using desiccated PIB as an edge sealer and POE as an encapsulant (see Figure 1A for

structure), three encapsulated perovskite minimodules were tested outdoors at the perovskite PV accelerator for commercializing technologies (PACT). The real-time minimodule temperature was recorded from the back side every 30 s (Note S1). As shown in Figure 1B, all of them exhibited obvious degradation after ~270 days of testing (occurred in summer), despite the fact that the highest minimodule temperature was below 50°C (Figure 1C). Dark lock-in thermography (DLIT) was employed to monitor changes in the electrical performance of the field perovskite minimodules. The DLIT image in Figure 1D reveals heating near the top and bottom Cu edges in the fielded module. As shown in Figure S1, the heating is co-located with bright electroluminescence (EL) intensity, indicating current crowding in these areas, which stems from nonuniform and higher series resistance in the other areas of the module. The high brightness and good uniformity of the photoluminescence (PL) image in Figure S2 for this field module suggest that the perovskite does not significantly degrade during outdoor tests.

We also did a study of perovskite minimodules, which are the sisters of those minimodules installed outdoors, after an indoor test of 1-sun illumination at 75°C for 100 h. The DLIT image (Figure S3) also exhibits increased heating along the scribe interconnects as well as near the edge of the Cu in the minimodule. Varying the current density in the DLIT images in Figures S4 and S5 demonstrates that the heat is generated from mild shunting along P3 as well as large-area series resistance issues, causing current crowding along the edges near the isolation scribes. These results further demonstrate that the P3 scribe lines serve as a pathway for the degradation of the perovskite minimodule, which is more susceptible to damage by external species such as oxygen or moisture.

State-of-the-art encapsulation cannot block oxygen

Consider that the encapsulation strategy we used has been frequently reported to effectively block moisture ingress and has resulted in a record outdoor operation lifetime.^{19,20} Other environmental factors (e.g., UV light and thermal cycling) are modest in this field test; we hypothesize that oxygen ingress will cause this degradation, as the PSC device is sensitive to oxygen and light.¹⁶ We first experimentally evaluated how fast oxygen diffuses through common perovskite minimodule encapsulation using PIB and POE. As illustrated in Figure 1A, a double-glass encapsulation structure consisting of a PIB edge seal and a POE encapsulant was used, each ~500 μm thick after lamination. We then extrapolated the oxygen diffusivity in PIB and POE at 70°C using Arrhenius fitting of experimentally measured diffusivity at other temperatures from the literature (Figure S6; Table S1).^{21,22} As summarized in Figure 2A, the diffusivity of oxygen in PIB ($9.42 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$) is 27% of that in POE ($3.48 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$) at 70°C. Oxygen diffusion through 1-cm-wide PIB and 5-cm-wide POE slabs was simulated at 70°C using FiPy Grid1D mesh (details in Note S2).^{23,24} As shown by the simulation result in Figure 2B, it takes only 1 day for oxygen to diffuse through a 1-cm-wide PIB edge sealer, and 11 days for oxygen to diffuse through 5-cm POE at 70°C, which is consistent with the previous report.¹⁸ Therefore, the perovskite solar minimodules with an area of 100 cm² or less would be quickly saturated with oxygen in the POE encapsulant at 70°C. For large

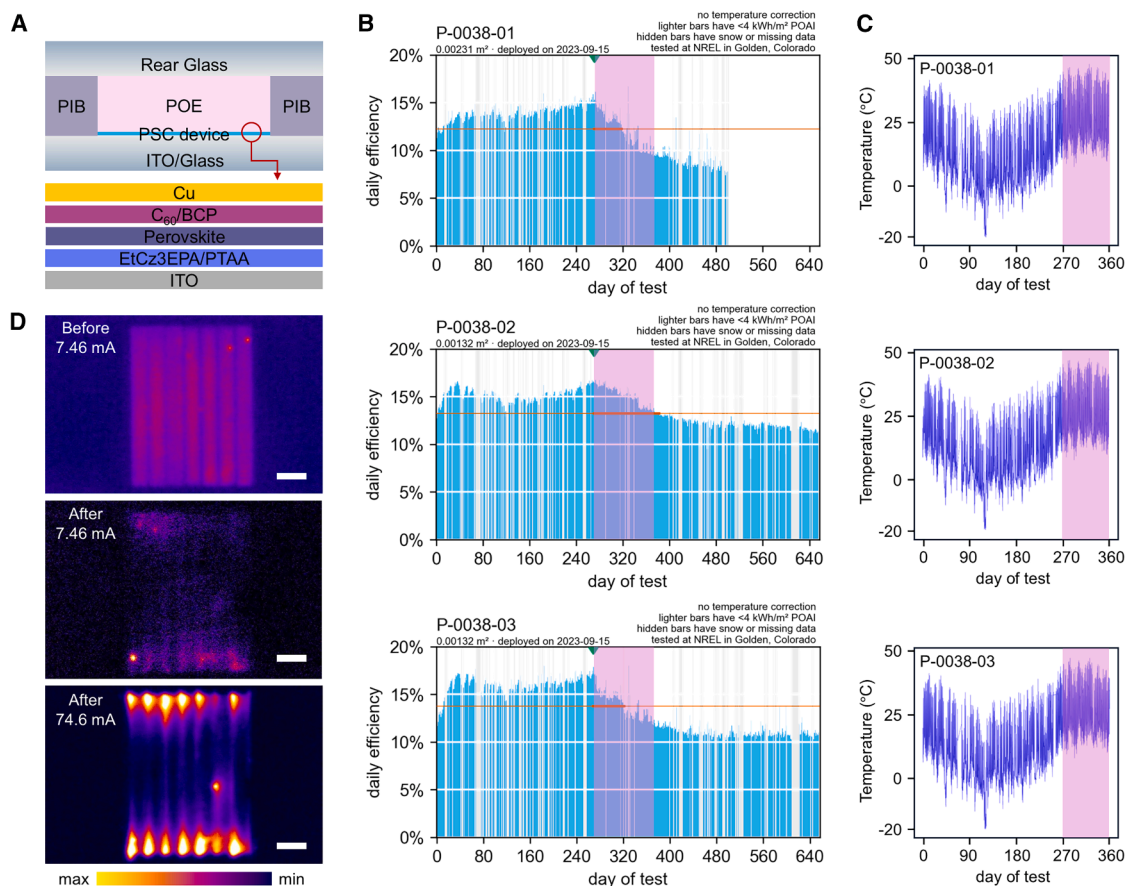


Figure 1. Unexpected degradation of the perovskite minimodule in the outdoors

(A) The structure of an encapsulated perovskite minimodule by PIB and POE, the enlarged part is the configuration of the PSC device. The image is not plotted in scale for clarity.

(B) The outdoor performance of the PSC minimodule was tested at PACT. The missing data for P-0038-01 was due to being removed from the test earlier.

(C) The back side hourly temperature of the fielded minimodule post-processed from the recorded temperature at PACT.

(D) The DLIT image (scale bars represent 1 cm) of the field perovskite minimodule (P-0038-01) before and after outdoor test. The bright region in the DLIT image represents heating of the device under forward bias at currents of 7.46 mA and 74.6 mA.

modules with an area of $1 \times 2 \text{ m}^2$, although it takes about 1 year for oxygen to reach the middle of the center, the possible damage to the subcells at the periphery of the solar panels could dramatically reduce their efficiency, due to serialization of the cells in perovskite modules.

The oxygen solubility (Figure S7; Note S3) in PIB ($4.68 \times 10^{-5} \text{ mol m}^{-3} \text{ Pa}^{-1}$) is >2 times higher than that in POE ($1.87 \times 10^{-5} \text{ mol m}^{-3} \text{ Pa}^{-1}$).^{21,22} The saturated oxygen concentrations in PIB and POE are calculated to be 1.0 mol m^{-3} and 0.4 mol m^{-3} at 70°C , respectively. These oxygen concentrations correspond to roughly 5%–15% of that in air, but oxygen can continuously permeate into the POE once it is consumed by its reaction within the PSCs. Given that the oxygen diffusivity in both PIB and POE follows the Arrhenius relationship, oxygen diffusion was simulated using the hourly measured minimodule temperature data. Here, we simulated oxygen diffusion across 1-cm PIB and 1-cm POE, corresponding to the potential degradation at the first subcells. After exposure to air for 270 days, the oxygen within 1-cm PIB

reaches almost saturation (Figure 2C), whereas the oxygen concentration at POE 1-cm from the PIB is $\sim 0.3 \text{ mol m}^{-3}$. This indicates that oxygen exposure is a plausible contributor to the observed minimodule degradation.

To verify these calculations, we measured the actual oxygen concentration in encapsulated mock perovskite minimodules after oxygen exposure using a device structure illustrated in Figure 2D. The mock minimodules have the same structure as working perovskite minimodules, while the perovskite in the central area was removed and replaced with an oxygen sensor film. Figure 2E shows the image of a sensor film integrated into a perovskite minimodule. Here, we selected platinum(II) meso-tetra(pentafluorophenyl)porphyrine (PTTFPP) as the oxygen-sensing material whose chemical structure is shown in Figure S8. PTTFPP was reported to have high sensitivity to oxygen with a lowest detection limit of $3 \times 10^{-4} \text{ mol m}^{-3}$.^{25,26} PTTFPP was mixed in polymethyl methacrylate (PMMA) to disperse PTTFPP for optimized thin film coating. The sensor film thickness was controlled below $1 \text{ }\mu\text{m}$,

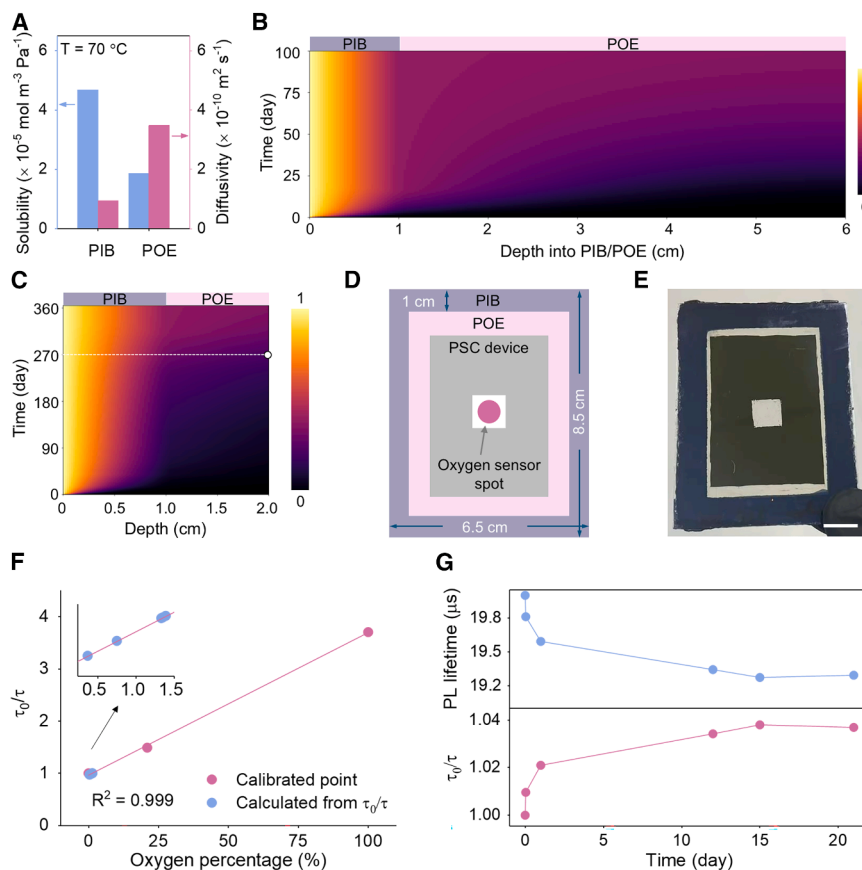


Figure 2. PIB and POE are not sufficient to block oxygen

(A) The calculated solubility (blue) and diffusivity (purple) of oxygen in PIB and POE at 70°C, respectively.

(B) Simulated concentration of oxygen as a function of depth and time in 1 cm PIB with 5 cm POE at 70°C.

(C) Simulated concentration of oxygen as a function of depth and time in 1 cm PIB with 1 cm POE under the hourly temperature for P-0038-01. The color bar is the oxygen concentration (mol m^{-3}). (D and E) The scheme (D) and a photograph (E, scale bars represent 1 cm) of a custom-fabricated oxygen sensor material integrated at the center of a PIB and POE-encapsulated minimodule (6.5 cm $\times$ 8.5 cm).

(F) The calibration of the oxygen sensor film and calculated oxygen percentage in encapsulated PSC minimodules using the τ_0/τ value. The inset shows the low oxygen percentage region (<1.5%). (G) PL lifetime evolution and τ_0/τ of the oxygen sensor integrated in the encapsulated PSC minimodule placed at 70°C in the dark for over 20 days.

matching that of perovskites, which avoids stress on the neighboring POE during encapsulation and ensures unchanged oxygen diffusion through POE. The PL lifetime of PTTFPP/PMMA films under different oxygen concentrations was first recorded, as presented in Figure S9, reducing from 20.0 μs in pure nitrogen to 5.4 μs in pure oxygen. As shown in Figure 2F, the inverse of PL lifetime of the sensor film is proportional to the oxygen concentration at the entire concentration range from 0 to 100%, which can be described by the Stern-Volmer quenching equation²⁵:

$$\frac{\tau_0}{\tau} = 1 + K_{\text{SV}}[\text{O}_2],$$

where τ_0 and τ are the PL lifetimes of the sensor films in nitrogen and different oxygen percentages, respectively.

After encapsulation, the PL lifetime was slightly reduced to 19.8 μs (Figure S10), corresponding to the 0.15 mol m^{-3} oxygen remaining in POE calculated by τ_0/τ in Figure 2G. It indicates the presence of some residual oxygen in POE, despite the fact that both PIB and POE films were pretreated at 50°C in a nitrogen glovebox for 12 h to remove dissolved oxygen. During the storage of the dummy minimodule in the dark at 70°C in air, the PL lifetime from the oxygen sensor film continued to reduce and stabilized at ~19.3 μs after 12 days (Figure 2G). This duration is close to the simulated value to reach saturation for oxygen to diffuse through a

PIB and POE, which could damage the perovskites or other layers.

C₆₀ oxidation as the dominant degradation pathway

We further evaluated the impact of oxygen on the durability of the p-i-n structured PSCs. Thermally evaporated C₆₀ is widely used as an ETL in inverted PSCs and perovskite modules due to its excellent electron transport capability, relatively low cost, reducing photocurrent hysteresis, and reasonably good thermal stability.^{27–29} We used a setup shown in Figure S11 to measure stability of these unencapsulated PSCs under illumination from a metal halide lamp (1 sun intensity, 4.5% UV light) at 70°C, with varying oxygen concentrations. As shown in Figure 3A, the efficiency of such devices after oxygen exposure for 2 h decreased by 35% with an obvious reduction in fill factor (FF) and short-circuit current density (J_{SC}). The open-circuit voltage (V_{OC}) of the devices reduced slightly by ~15 mV (inset in Figure 3A). The huge loss of FF and J_{SC} likely indicates a charge-extraction barrier formed in these devices. Statistical data from 12 individual devices in Figure 3B indicate that the reduction in PCE primarily took place during the initial 2 h (~35%), with only a slight reduction observed over the subsequent 46 h.

Since C₆₀ is a top layer that is exposed to oxygen through a thin layer of Cu, we speculate that C₆₀ was damaged during the oxygen exposure. Figure S12 shows that the vertical

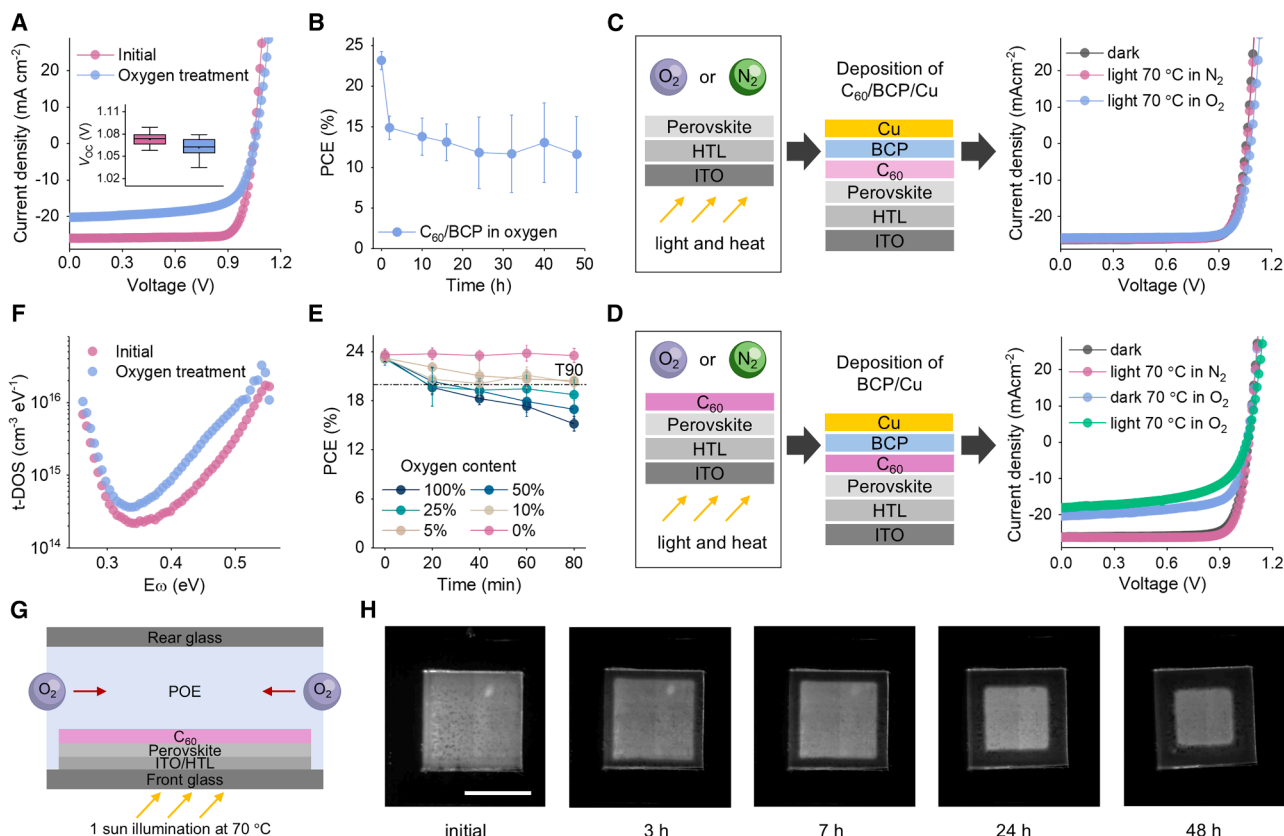


Figure 3. C_{60} oxidation decreased the efficiency of the PSCs

(A) The J-V curves of an unencapsulated C_{60} /BCP-based device before and after oxygen treatment for 2 h under 1-sun illumination at 70°C. The inset shows the statistical data of V_{OC} before and after oxygen treatment (data collected from 12 devices).
 (B) The PCE evolution of the C_{60} /BCP-based device under 1-sun illumination at 70°C in oxygen for 2 days (data collected from 12 individual devices).
 (C) Schematic illustration (left) and J-V curves (right) for pretreated bare perovskite film with oxygen or nitrogen under 1-sun illumination at 70°C for 1.5 h before fresh C_{60} deposition. RT, room temperature.
 (D) Schematic illustration (left) and J-V curves (right) for pretreated perovskite with C_{60} layer under oxygen or nitrogen, 1-sun illumination, and 70°C for 1.5 h before fresh BCP deposition.
 (E) The PCE evolution of a nonencapsulated C_{60} /BCP-based device under 1-sun illumination at 70°C with different oxygen contents (0% means nitrogen, data from 9 devices).
 (F) The initial and oxygen-treated (2 h) t-DOS curve of the C_{60} /BCP-based device.
 (G) Schematic illustration of the oxygen diffusion across the POE-encapsulated perovskite film under 1-sun illumination at 70°C supplied from the front side. The top surface of the perovskite film was encapsulated using a ~ 500 μm POE layer laminated with a rear glass substrate.
 (H) The PL mapping from the ITO sides of a POE-encapsulated perovskite film with C_{60} (structure is ITO/HTL/perovskite/ C_{60} /POE/glass) before and after oxygen treatment (scale bars represent 1 cm).

conductivity of the C_{60} film (30 nm) was reduced by 1-fold under oxygen exposure for 2 h, consistent with the greatly decreased J_{SC} observed in PSC devices. To further verify this damage in C_{60} , half of the devices were fabricated and tested under the same conditions to find which layer was degraded. As shown in Figure 3C, we first stressed the stack of ITO/EtCz3EPA/PTAA/perovskite in oxygen, and then completed the device fabrication by evaporating C_{60} /BCP/Cu. Oxygen exposure of EtCz3EPA/PTAA/perovskite under 1-sun illumination at 70°C for 1.5 h slightly enhanced the PCE. It is in striking contrast to the reduced PCE of the stressed whole devices shown in Figure 3A, suggesting that the degradation caused by oxygen is not caused by oxidation of the perovskite layer. Then we added a C_{60} layer on top of the perovskite layer and stressed

them in oxygen before finishing device fabrication (Figure 3D). A significant PCE loss of 35% was observed after stressing the film stack with C_{60} , with a substantial reduction of J_{SC} and FF, resembling the result of stressing the completed devices. For a control study, we put the ITO/EtCz3EPA/PTAA/perovskite/ C_{60} in a nitrogen environment and tested them under the same conditions. In contrast, the devices made of such stressed films did not show any degradation, which again suggests that the degradation was primarily caused by the C_{60} oxidation.

The above results confirm that oxygen-induced PSC degradation is caused by damage to the C_{60} layer. To find out whether the oxidation of C_{60} -induced device efficiency reduction is a first-order reaction, i.e., proportional to oxygen concentration, we injected a mixed gas of nitrogen and oxygen into a chamber to

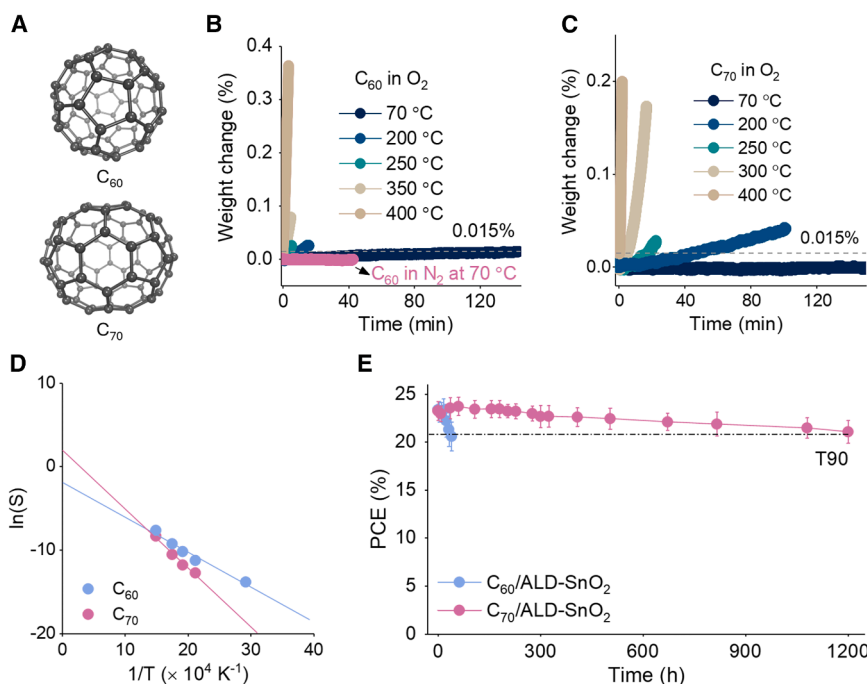


Figure 4. Oxygen-resistant C₇₀ slows down the degradation of PSCs

(A) The chemical structure of C₆₀ and C₇₀. (B and C) The oxygen adsorption experiment of C₆₀ powder (B) and C₇₀ powder (C) at different temperatures. (D) The relationship between ln(S) and 1/T for C₆₀ and C₇₀, where S represents the slope calculated from the time corresponding to a 0.015% weight increase at different temperatures, follows the Arrhenius relationship. (E) The PCE evolution of unencapsulated C₆₀/ALD SnO₂ (10 individual devices) and C₇₀/ALD SnO₂ (12 individual devices) based PSCs with 1-sun illumination at 70 °C in oxygen.

create an atmosphere containing different oxygen percentages and then placed PSCs in this chamber under 1-sun illumination at 70 °C (see Figure S13). As shown in Figure 3E, the device under nitrogen showed an unchanged PCE, whereas all unencapsulated PSCs with oxygen with different ratios degraded significantly in the initial 20 min, then the degradation slowed down. Thus, neither the efficiency loss of PSCs nor the C₆₀ oxidation rate follows first-order kinetics with oxygen partial pressure. Even low concentrations of oxygen lead to rapid, irreversible damage in both C₆₀ and PSCs.

Oxidation of C₆₀ could introduce many deep charge traps, limit charge extraction, and enhance non-radiative charge recombination, which causes the reduction of J_{SC} and V_{OC}. The increased series resistance of the PSC after oxidation (Figure S14) agrees with widespread current crowding observed in the EL of the field minimodule (Figure S1). Moreover, the thermal admittance spectroscopy (t-DOS) in Figure 3F revealed an increase of deep trap density (0.4–0.5 eV) by nearly one order of magnitude after stressing for only 2 h. The defects generated in C₆₀ can also cause PL quenching in perovskites. As shown in Figure 3G, a POE-encapsulated perovskite film with C₆₀ was used to evaluate the oxygen diffusion. Figure 3H presents the PL mapping of a perovskite/C₆₀ film encapsulated with POE between ITO glass and another piece of glass during oxygen treatment under 1-sun illumination at 70 °C, measured from the ITO glass side. A clear contrast of PL intensity can be observed between the regions with oxidized and pristine C₆₀. The PL pattern exhibited a ring-shaped pattern due to the permeation of oxygen from four edges, and the bright region kept shrinking during oxygen treatment with light at 70 °C, which directly visualizes the diffusion of oxygen in POE along the in-plane direction. Figure S15 shows the simulated oxygen diffusion process at 70 °C through POE using the

the Cu electrode acts as a barrier to delay the diffusion of oxygen into C₆₀.

C₇₀ greatly slows down the oxidation

Given that C₆₀ oxidation appears to drive degradation, using a fullerene with slower oxidation kinetics is expected to prolong the T90 of PSCs. Here, we selected C₇₀ with the chemical structure shown in Figure 4A, which has been broadly used for organic solar cells and is commercially available in large quantities. There are controversial results in the literature regarding the oxidation of C₇₀ compared with C₆₀. Earlier studies showed that C₇₀ has a larger oxidation potential in electrochemical measurement, indicating C₇₀ is harder to oxidize than C₆₀.³⁰ This was attributed to the less delocalized electrons of C₇₀ due to its elongated ellipsoid structure than the highly symmetrical spherical C₆₀.^{31–33} The elongated structure of C₇₀ results in a broader and more anisotropic electron distribution, whereas the uniform electron density and a higher π -electron delocalization of the symmetrical C₆₀ facilitate its reaction with oxygen. However, faster oxidation of C₇₀ than C₆₀ was reported at higher annealing temperatures of 250 °C–445 °C,³⁴ and the loss of second electrons was observed in C₇₀ but not in C₆₀.³⁰ Previous studies show that C₆₀ exposed to oxygen at 370 K forms two oxidation products (C₆₀>O and C₆₀=O), whereas C₇₀ predominantly forms only one product (C₇₀>O) with a smaller signal.³¹ These oxidation products in our experimental conditions should be the same due to the similar temperature used. The reduced oxidation of C₇₀ can be explained by the larger strain in C₆₀. The more distorted sp² bonds in the high-curvature C₆₀ make them easier to convert to sp³ hybrid carbon (C–O–C bond) upon oxygen exposure, because the reaction releases strain. The elongated structure in C₇₀ partially relieves strain and results in lower reactivity in the equatorial region with lower curvature.³⁵

To directly verify the chemical reaction of C_{60} with oxygen, we measured the weight change of C_{60} powder by exposing it to an oxygen environment at different temperatures using thermogravimetric analysis (TGA). We first conducted oxygen absorption experiments to track the weight change of C_{60} powder during oxygen exposure at 70°C. As shown in Figure 4B, the weight of C_{60} powder increased linearly with exposure time and increased by 0.015% after 145 min of oxygen exposure. The increased weight suggested that $\sim 3\%$ of C_{60} reacted with oxygen during this test. In striking contrast, the weight of C_{60} powder remained nearly unchanged when C_{60} powder was exposed to nitrogen. We then measured the weight change of C_{70} powder by employing the same method. As shown in Figure 4C, no obvious weight change of C_{70} powders was observed during the 2.5 h of oxygen exposure at 70°C, whereas the 0.015% weight increase of C_{70} was observed at 49 min, 19 min, 5.4 min, and 0.6 min at temperatures of 200°C, 250°C, 300°C, and 400°C, respectively. To understand the apparent reaction activation energy, we plotted the apparent rates (slope of weight change in TGA plots) of C_{70} and C_{60} powders versus the inverse of temperatures in Figure 4D. The data for both C_{70} and C_{60} follow the Arrhenius equation. The reaction energy barrier of C_{70} is fitted to 0.61 eV, which is nearly twice that of C_{60} (0.37 eV). The time required for C_{70} to reach a 0.015% weight increase at 70°C and 50°C was extrapolated to be 337 h and 1,065 h, respectively, which are 107 and 151 times longer than those of C_{60} . These observations confirm that the elongated structure of C_{70} , with its broader and more anisotropic electron distribution, decreases the reactivity between oxygen and the fullerene.

Subsequently, C_{70} was used to replace C_{60} to fabricate PSCs with a structure of ITO/EtCz3EPA/PTAA/perovskite/ C_{70} /BCP/Cu. The same fullerene thickness of 30 nm was used. Replacing C_{60} with C_{70} did not change the efficiency of PSCs (Figure S17) despite their slightly different energy levels. The C_{70} /BCP-based PSCs retained over 90% of their initial PCEs after being illuminated by 1 sun at 70°C in oxygen for 64 h (Figure S18), which is 213-fold longer than the C_{60} /BCP device tested under the same conditions. This increase is consistent with the increased oxygen resistance of C_{70} measured by TGA. To understand why the more oxidation-resistant C_{70} still yielded only a T90 of 64 h, we analyzed oxygen diffusion through the BCP layer, using ALD-SnO₂, which is widely used in PSCs to enhance the thermal stability, reverse-bias stability, and operational stability, as a benchmark for its highly compact property.^{11,36–39} We quantified the oxygen diffusion rate across BCP and ALD-SnO₂ by using oxygen sensor films and confirmed the excellent oxygen-blocking effect of ALD-SnO₂ (Figures S19–S21). Therefore, ALD-SnO₂ was used to replace BCP, and the structure is thus changed to ITO/EtCz3EPA/PTAA/perovskite/ C_{60} /ALD-SnO₂/Cu. As presented by Figure 4E, even with nearly two orders of magnitude lower oxygen diffusion, oxygen still caused degradation of the C_{60} /ALD-SnO₂-based devices with a T90 of only 40 h (*J-V* shown in Figure S22). This result exhibits that ALD-SnO₂ alone is not sufficient, as it physically slows down the oxygen diffusion but cannot solve the intrinsic oxidative vulnerability of C_{60} . The delayed oxygen penetration thus represents a transient mitigation rather than a permanent solution.

We therefore combined C_{70} and ALD-SnO₂ to improve the oxygen resistance of PSCs with a structure of ITO/EtCz3EPA/PTAA/perovskite/ C_{70} /ALD-SnO₂/Cu. As shown in Figure 4E, the unencapsulated PSCs employing an ETL of C_{70} /ALD-SnO₂ retained 90% of their initial PCE after exposure to 1 atm oxygen at 70°C for 1,200 h, 30 times enhancement to the C_{60} /ALD-SnO₂-based device. The *J-V* curves of a typical device are shown in Figure S23 before and after exposure for 1,200 h. The PCE reduction of the aged devices came mainly from a slight reduction in J_{SC} , and almost unchanged FF and V_{OC} . This PCE reduction is not dominated by heating at 70°C or illumination, because the control PSCs with C_{70} and ALD-SnO₂ retained 95.2% of their initial PCE after testing for 1,440 h under 1-sun illumination at 70°C in nitrogen (Figure S24). This 5% lifetime difference for the devices tested in nitrogen and oxygen can be attributed to the slow oxidation of C_{70} by oxygen at 70°C.

Increased minimodule stability by C_{70}

Finally, we continued to evaluate the operational stability under elevated temperature of unencapsulated perovskite minimodules. Two types of perovskite minimodules were fabricated by blade coating and applied to evaluate their impact on C_{70} . Figure S25 presents the *I-V* curves of encapsulated FA_{0.9}CS_{0.1}PbI₃-based minimodules with C_{60} or C_{70} as ETLs, both exhibiting comparable PCEs exceeding 19% with an aperture area of 18 cm². It is noted that no anti-reflection coating was applied in this study, which generally enhances the minimodule efficiency by 0.5%–1%. Then, the minimodule stability under 1-sun illumination at 70°C in air was tracked. As shown in Figure 5A, minimodules with C_{70} /ALD-SnO₂ retained 94% of their initial PCE after aging for 1,512 h, whereas the C_{60} /BCP-based minimodules lost 44% of their initial PCE. Both types of perovskite minimodules exhibited a slightly longer lifetime than the corresponding small-area devices tested by the same method, which may be attributed to the wider Cu electrode in these minimodules, which slows down the oxygen lateral diffusion into fullerene. In addition, in contrast to the nearly linear degradation observed for the small-area devices, the minimodule degradation is clearly nonlinear, which may be explained by the fact that the initial degradation of peripheral subcells limits the minimodule efficiency.

Another perovskite with a nominal composition of Cs_{0.04}MA_{0.14}FA_{0.82}Pb(I_{0.987}Br_{0.013})₃ was further employed to evaluate the universality of C_{70} in improving oxygen resistance, and the perovskite minimodules were fabricated as described elsewhere.²⁹ The PCE of minimodules (6 cm²) with C_{70} /ALD-SnO₂ as ETL is 22.2%, comparable with the C_{60} /ALD-SnO₂-based minimodule (22.5%, *J-V* curve in Figure 5B). As shown in Figure 5C, the unencapsulated perovskite minimodules employing C_{70} /ALD-SnO₂ as ETL retained 90% of their initial PCE after 500 h under 1-sun illumination at 70°C in oxygen, outperforming the corresponding minimodules with C_{60} /ALD-SnO₂ (retained 54%). Again, to benchmark the stability of the minimodules with this perovskite composition, we measured the stability of these minimodules under 1-sun illumination at 70°C. All these minimodules retained over 90% after light soaking at 70°C (Figure S26) for 500 h in nitrogen, demonstrating that

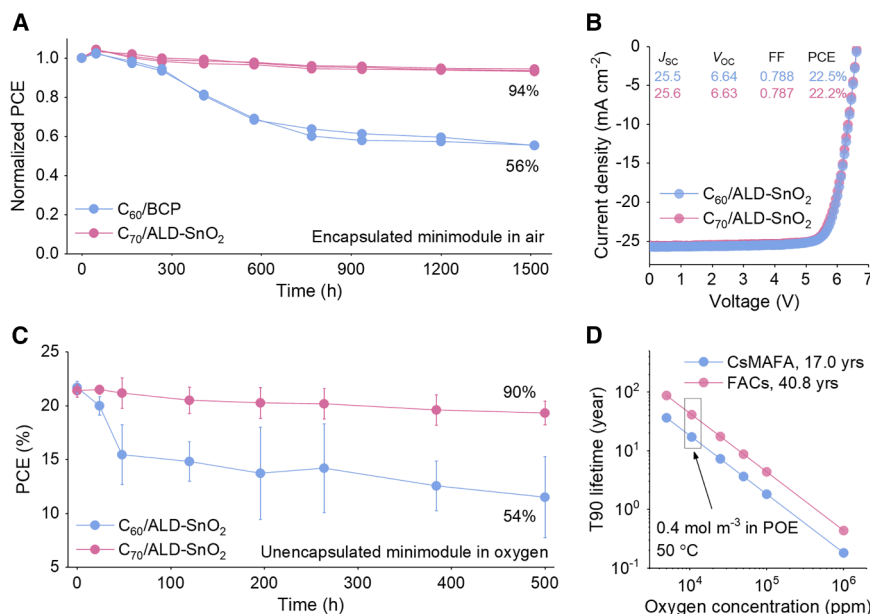


Figure 5. Combining C_{70} and ALD- SnO_2 increased the minimodule stability in oxygen

(A) The normalized PCE evolution of encapsulated C_{60}/BCP (2 individual devices) and $C_{70}/ALD-SnO_2$ (3 individual devices) based perovskite minimodule under 1-sun illumination at 70°C in air (35%–45% humidity).

(B) The J-V curve of $C_{60}/ALD-SnO_2$ and $C_{70}/ALD-SnO_2$ -based perovskite minimodule with 6 sub-cells.

(C) The PCE evolution of unencapsulated $C_{60}/ALD-SnO_2$ (5 individual devices) and $C_{70}/ALD-SnO_2$ (10 individual devices) based perovskite minimodule under 1-sun illumination at 70°C in oxygen.

(D) The estimated T90 lifetimes of the two types of perovskite composition ($FA_{0.9}Cs_{0.1}PbI_3$, FACS; $Cs_{0.04}MA_{0.14}FA_{0.82}Pb(I_{0.987}Br_{0.013})_3$, CsMAFA) as a function of oxygen concentration (0.4 mol m^{-3} corresponding to the stabilized oxygen in POE) at 50°C, followed by the first-order kinetics and Arrhenius equation.

oxygen is the main stressor for the decreased efficiency of the C_{60} -based minimodules.

We extrapolate the lifetime of the perovskite minimodules based on these accelerated testing results. The oxygen concentration within encapsulated perovskite minimodules is determined by the solubility of oxygen in POE. Assuming the perovskite solar minimodule degradation is solely caused by oxidation, which follows a linear relationship between oxygen concentration and reaction rate, we derive a T90 lifetime for the encapsulated perovskite minimodules operating at 50°C (Figure 5D). This extrapolation neglects other extrinsic field stressors (e.g., bias and mechanical fatigue) and thus represents an upper-bound lifetime under oxygen-limited encapsulated conditions. The T90 lifetimes of encapsulated $FA_{0.9}Cs_{0.1}PbI_3$ and $Cs_{0.04}MA_{0.14}FA_{0.82}Pb(I_{0.987}Br_{0.013})_3$ perovskite minimodules at 50°C are extrapolated to be 40.8 years and 17.0 years, respectively, by using the Arrhenius equation with the measured activation energy.

DISCUSSION

In summary, we first experimentally confirmed that oxygen could diffuse through PIB and POE quickly, exposing a high concentration of oxygen to PSCs. Unencapsulated PSCs with C_{60} quickly degraded in pure oxygen under 1-sun illumination at 70°C with a T90 of only 0.3 h, and the degradation was found to be caused by C_{60} oxidation. We addressed this issue by replacing C_{60} with C_{70} , yielding extrapolated T90 lifetimes of the perovskite solar minimodules to be 17–41 years. This simple approach does not affect the established manufacturing process for perovskite minimodules, and C_{70} is an abundant material with a well-established synthesis process, making it easy to be adapted by the industry. These findings highlight that choosing highly oxidation-resistant ETLs (such as C_{70}) is crucial for achieving stable perovskite minimodule operation in ambient air.

METHODS

Materials

2-methoxyethanol (2ME, 99.9%), dimethyl sulfoxide (DMSO, 99.9%), toluene (99.9%), tetrahydrofuran (THF), and methanol were purchased from Sigma-Aldrich. Lead iodide (PbI_2 , 99.999%) was purchased from Advanced Election Technology Co., Ltd. FAH_2PO_2 was synthesized in our lab according to the previous report.⁴⁰ Formamidinium iodide (FAI, >99.99%), formamidinium chloride (FACl, >99.99%), and phenylammonium tetrafluoroborate ($PEABF_4$) were purchased from GreatCell Solar. Cesium iodide (CsI, 99.999%), zinc chloride ($ZnCl_2$, >97%), carbonyldiimidazole (CDI, 98%), PTAA (average Mn 7,000–10,000), and BCP (96%) were purchased from Sigma-Aldrich. C_{60} and C_{70} were purchased from Nano-C Inc and Xiamen Funano New Material Technology Co., Ltd. EtCz3EPA was synthesized by Alan Sellinger's Group.¹⁰ PTFPP (>95%) was purchased from Frontier Specialty Chemicals.

Device fabrication

The ITO substrates were pre-patterned and cleaned with acetone under ultrasonic instruments. All substrates were treated by UV-ozone for 20 min before use. The perovskite layer and hole transporting layer were prepared by blade coating at a speed of 20 mm s^{-1} in an air chamber with a humidity controller (20°C, 20%–40% RH).

For the EtCz3EPA/PTAA bilayer: EtCz3EPA (1 mg mL^{-1} in methanol) was blade-coated on the ITO substrate and heated at 150°C for 3–5 min, washed with methanol to form a self-assembled monolayer. PTAA (2.2 mg mL^{-1} in toluene) was deposited on the top of EtCz3EPA without heating.

For the deposition of perovskites, $1 \text{ M } FA_{0.9}Cs_{0.1}PbI_3$ was prepared by mixing $FAPbI_3$ (2.2 M in 2ME), $CsPbI_3$ (2.2 M in DMSO), FAH_2PO_2 (5 wt% in 2ME), FACl (50 mg mL^{-1} in 2ME), $PEABF_4$

(10 mg mL⁻¹ in 2ME), CsI (0.12 M in 2ME), ZnCl₂ (0.12 M in 2ME) and CBH (10 mg mL⁻¹ in 2ME). The perovskite precursor was blade-coated on top of the hole transporting layer (HTL) with a gap of 350 μm at a speed of 20 mm s⁻¹. The wet perovskite film was heated at 150°C for 3.5 min and 100°C for 2.5 min. The thickness of the perovskite film is 800–1,000 nm.

Then, the perovskite film was deposited by thermal evaporation of C₆₀ or C₇₀, BCP (6 nm) or ALD-SnO₂ (10–15 nm), and Cu. For a 200-cycle ALD-SnO₂ sample, the pulse of the tin source and water were 0.33 s and 0.015 s.

For the minimodule fabrication, the encapsulated perovskite minimodules were fabricated on pre-patterned large ITO glass substrates with a P1 width of 30 μm, followed by the same procedure as the small PSCs. The area of each subcell is 3.575 cm². The width of each subcell is 6.5 mm, whereas the final P2 and P3 widths were measured to be ~120 μm and ~100 μm, respectively. The total width of the dead area was measured to be ~350 μm, giving a geometric fill factor (GFF) of 94.6%.

The encapsulation process for the module: the integrated oxygen sensor-based mock PSC minimodule or minimodule operated outdoors and stability test was encapsulated by desiccated PIB (edge sealer, Quanex, SET LP03) and POE (encapsulant, Dow) with a laminator (LAB-LAMINATOR L036LAB) at 110°C for 10 min.

For another perovskite with a nominal composition Cs_{0.04}MA_{0.14}FA_{0.82}Pb(I_{0.987}Br_{0.013})₃, the perovskite precursors were prepared by dissolving MABr (9 mg), PbCl₂ (22 mg), CsI (24 mg), MAI (47.6 mg), FAI (292.4 mg), and PbI₂ (968 mg) in DMF solvent with further addition of DMSO and NMP. Prior to use, the precursor solution was shaken by a vortex machine until it was fully dissolved. The HTL was based on a self-assembled monolayer of mixed solution MeO-2PACz and Me-4PACz (0.5 mg mL⁻¹ in ethanol, volume ratio 3:2). The minimodules were fabricated on pre-patterned large fluorine doped tin oxide (FTO) glass substrates with a P1 width of 30 μm. The width of each subcell is 3.3 mm, whereas the final P2 and P3 widths were measured to be 70 μm and 45 μm, respectively. The total width of the dead area was measured to be ~200 μm, giving a GFF of 94%. The aperture area of minimodules (with 6 subcells) is 6 cm².

Device characterization

Photocurrent measurement

The *J*-*V* curve of the PSC device was tested by Newport-91150V-KG5 (Oriol Sol3A, Class AAA Solar Simulator), a Xenon lamp-based solar simulator. The light intensity of the illuminator was calibrated to 100 mW cm⁻² by a reference silicon cell. All the data was recorded by Keithley 2400 source meters in air and at room temperature. The *J*-*V* curves were obtained from a reverse scan (1.2 V to -0.1 V) with a scanning rate of 0.01 V s⁻¹ and sweep delay of 10 ms.

Isothermal experiment

The isothermal experiment was conducted by fixing the temperature and flowing with dry oxygen or nitrogen to see whether any weight changes occurred in C₆₀ or C₇₀ powder. The measurement was conducted by a Thermogravimetric Analyzer (TGA), TGA5500.

TRPL test

Transient photoluminescence (TRPL) measurement was conducted on a DeltaFlex modular fluorescence lifetime system

from HORIBA Scientific with a pulsed laser source at 405 nm (HORIBA NanoLED 402-LH, ~1 mm² spot size). The PL signal at 650 nm was collected by using time-corrected single-photon counting.

t-DOS measurement

t-DOS measurement was performed by thermal admittance spectroscopy using an LCR meter (Agilent E4980A). The tDOS was derived from the frequency-dependent capacitance and voltage-dependent capacitance based on the equation:

$$N_T(E_\omega) = -\frac{1}{qkT} \frac{\omega dC}{d\omega} \frac{V_{bi}}{W},$$

where *q*, *k*, *T*, *ω*, and *C* are the elementary charge, Boltzmann's constant, temperature, angular frequency, and specific capacitance, respectively. *W* and *V_{bi}* are the depletion width and built-in potential, respectively.

PL mapping

PL mapping was conducted with the PicoQuant MicroTime 100 and FluoTime 100 systems at room temperature. The wavelength of excitation is 365 nm.

The stability test

All the small-area devices were tested with 3M tape covering the active area, avoiding the scratching of the electrode caused by movement. The metal halide lamp containing 4.5% UV light was used as a light source, and the temperature was set at 70°C. For the oxygen flow experiment, the flow rate of oxygen was set as 2 psi. The chamber for oxygen flow and stability test was purchased from Wuhan 91PVKSolar Technology Co., Ltd, China.

Visualization of chemical structure

VMD was used for the visualization of the chemical structure of C₆₀ and C₇₀.⁴¹

Basis for diffusivity values

Accurate modeling of oxygen diffusion into polymer encapsulants requires reliable estimates of oxygen solubility (*S*) and diffusivity (*D*) across relevant temperatures (-20°C~110°C). However, direct measurements for PIB and POE at desired conditions were limited in the available literature.¹⁸ To address this, we extrapolated the temperature dependence of *D* by using the Arrhenius diffusivity equation:

$$D(T) = D_0 \exp\left(-\frac{E_a}{RT}\right),$$

where

- *D*₀ is the pre-exponential factor (m² s⁻¹),
- *E_a* is the activation energy for diffusion (J mol⁻¹),
- *R* is the universal gas constant (8.314 J mol⁻¹·K⁻¹),
- *T* is absolute temperature (K).

Experimentally measured values of *D* at lower temperatures (e.g., 17°C–50°C) were extracted from Van Amerongen (for PIB)²¹ and Seguchi et al. (for POE).²² These data points were plotted as ln(*D*) versus 1/*T*, producing a straight line according to the linearized Arrhenius form:

$$\ln(D) = \ln(D_0) - \frac{E_a}{RT}.$$

A linear regression was performed to obtain the slope and intercept, from which E_a and D_0 were determined.

Basis for solubility values

Unlike diffusivity, solubility does not fundamentally follow an Arrhenius-type temperature dependence, as it lacks a universally accepted governing equation tied to energy barriers. Therefore, no direct exponential model was applied to solubility. Instead, solubility values were estimated using linear interpolation and extrapolation across temperature, assuming no phase transitions or structural changes in the polymer over the studied range (17°C–110°C). While solubility can be expressed as the ratio $S = Q/D$, and both Q and D individually follow Arrhenius behavior, the resulting temperature dependence for S is inherited and does not reflect a true thermodynamic Arrhenius relationship.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Jinsong Huang (jhuang@unc.edu).

Materials availability

This study did not generate new, unique materials. All of these materials are commercially available.

Data and code availability

All data are available in the main text or the [supplemental information](#).

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AUTHOR CONTRIBUTIONS

J.H. conceived the idea. H. Guo designed, fabricated, and characterized perovskite devices. A.N.S. simulated and visualized the oxygen diffusion in PIB and POE. H.W. did the PL mapping for the perovskite film. X.S. and C.F. did the optimization of EtCz3EPA/PTAA bilayer. C.G. conducted the AFM measurement. H. Guo, C.F., and M.W. optimized the perovskite recipe. H. Guo and X.S. did ALD-SnO₂ optimization. H. Guo, H.W., and C.G. discussed the design of oxygen sensor experiments. M.W. did the t-DOS measurement. H.J. optimized the encapsulation process of the perovskite module. M.O.-B., T.J.S., and L.T.S. conducted outdoor tests for the perovskite minimodules and did the analysis. J.W.S. and D.B.K. analyzed the PL, EL, and DLIT images for the field perovskite minimodule. A.L., B.D.D., and K.Z. fabricated the perovskite minimodule with Cs_{0.04}MA_{0.14}FA_{0.82}Pb(I_{0.987}Br_{0.013})₃ for testing. H. Guo and J.H. wrote the manuscript, and all authors commented on the manuscript.

DECLARATION OF INTERESTS

H.G., J.H., B.D.D., and A.L. are authors of a provisional patent (application no. 63/924,352) based on this manuscript. B.D.D. is the director of research and development at CubicPV Inc. B.D.D. and A.L. are employed by CubicPV Inc. J.H. is the founder of Perotech Inc. These relationships have been disclosed to and are under management by UNC-Chapel Hill.

SUPPLEMENTAL INFORMATION

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